NextAmp SIA

Target Azure application architecture — current scope + future enhancements

Current scope · New / changed in this version · Future · Cost optimization

USERS & ACCESS

Agent / Broker Portal
Web browser

Admin / Underwriter Portal
Web browser

Third-party systems
Inbound API calls (HTTPS)

HTTPS

EDGE & ENTRY (PUBLIC)

Azure Front Door + WAF
Single public entry · WAF · Standard (lean) / Premium (hardened)

Azure Static Web Apps
Agent and admin portals

Azure API Management
v2 tier · auth, rate limits, routing

SHARED PLATFORM (ACROSS CLIENTS)

Microsoft Entra ID
SSO / authentication

Azure Container Registry
Container images

REST / HTTPS

PER-CLIENT AZURE SUBSCRIPTION · APPLICATION CORE (SINGLE-TENANT)

AZURE CONTAINER APPS ENVIRONMENT (VNET-INTEGRATED)

Submission API Service

Intake / Core Service
Configuration & parties

Identity & Auth Service
Roles & access (uses Entra ID)

Intelligence Service (API)

Intelligence Worker
Async processing

Audit / Logging Service

Integration Service
Outbound connectors

HTTPS / APIs

Azure Service Bus (topics & queues)
Event-driven messaging · Standard (lean) / Premium (private endpoints)

DATA & AI (PRIVATE ENDPOINTS ONLY)

Azure Database for PostgreSQL
Application data · point-in-time restore + geo-backup

Azure Blob Storage
Documents / files

Azure OpenAI / AI Services
Content understanding

Azure Key Vault (per client)
Secrets & certificates

OBSERVABILITY (CENTRALIZED)

Azure Monitor + Application Insights + Log Analytics
Logs, metrics, traces, alerts

Private networking
VNet integration + private endpoints

Egress control
NAT Gateway (lean) / Azure Firewall (hardened)

Managed identity
Least-privilege access

Secure by default
Encryption at rest and in transit

NSGs on every subnet
Deny by default, allow only needed flows

Private DNS zones
Resolve private endpoints internally

Defender for Cloud + Policy
Continuous compliance checks

Token validation
Entra ID JWT at APIM + RBAC in services

EXTERNAL INTEGRATIONS

Customer systems
CRM, policy, etc.

External data providers
Credit, KYC, etc.

Microsoft 365
Outlook / Graph API

Power Automate
Workflow automation

COST OPTIMIZATION (BUILT IN)

Scale to zero
Container Apps Consumption plan for workers and quiet services

Lean vs hardened tiers
Start lean; upgrade SKUs only when a client requires it

Non-prod savings
Burstable PostgreSQL, off-hours shutdown, smaller SKUs

Storage lifecycle
Blob moves to Cool / Archive as documents age

Telemetry and AI usage
Log caps, trace sampling, token limits, smaller models

FinOps guardrails
Tags per client, budgets + alerts, reserved capacity for Prod

FUTURE ENHANCEMENTS (PLANNED)

Terraform (IaC)
Reusable, repeatable deployments

Dev / UAT / Prod
Multi-environment rollout per client

GitHub Actions CI/CD
Automated build, test, deploy

Cost governance
Monitoring, alerts, right-sizing

Backup & DR plan
Region failover · define RPO / RTO

Azure Cache for Redis
Only if performance needs it

KEY DESIGN GOALS

Secure by default
Zero trust, least privilege

Scalable microservices
Independent, resilient

Event-driven processing
Asynchronous, loosely coupled

Dedicated per-client deployment
Single-tenant architecture

Centralized observability
Logs, metrics, traces